



ICAIN-2026
International Conference on Artificial
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******* CALL FOR PAPERS *******

SPECIAL SESSION ON

Quantum-Inspired Optimization of Neuromorphic Self-Supervised Heterogeneous Graph Transformers for Multimodal Neurodegenerative Disease Analysis

SESSION ORGANIZERS:

[1. Dr. Tarak Hussain

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EDITORIAL BOARD: (Optional)

[Name, University or Organization, Country, e-mail]

SESSION DESCRIPTION:

Neurodegenerative disorders such as Alzheimer's, Parkinson's, and Huntington's diseases require a comprehensive analysis of heterogeneous multimodal clinical, neuroimaging, genetic, and longitudinal data. Traditional AI methods are still hampered by dependence on annotation, high processing cost, and insufficient modeling of complicated cross-modal and temporal interactions.

This special session will focus on the use of quantum-inspired optimization, neuromorphic intelligence, self-supervised learning, heterogeneous graph transformers and multimodal AI for the investigation of neurodegenerative illness. It offers optimal feature and architectural search, energy-efficient neuromorphic learning, label-efficient representation learning, and graph-based modeling of the patient, brain, biomarker, and clinical linkages.

The session invites submissions of research on explainable and trustworthy AI, federated learning, domain adaptation, uncertainty quantification, multimodal foundation models, cross-cohort generalization, and clinically deployable AI, to advance next-generation computational methods for neurodegenerative disease diagnosis, prognosis, progression modeling, and personalized healthcare.

RECOMMENDED TOPICS:

Topics to be discussed in this special session include (but are not limited to) the following:

- Quantum-inspired optimization for neuromorphic medical AI
- Self-supervised heterogeneous graph transformers for neuroimaging
- Spiking graph transformers for multimodal brain representation learning
- Quantum-inspired multimodal feature and architecture optimization
- Dynamic brain-clinical knowledge graphs for disease progression
- Neuromorphic multimodal fusion for neurodegenerative disease diagnosis
- Explainable graph-transformer models for neurological biomarkers
- Federated and privacy-preserving multimodal neurodegenerative AI
- Energy-efficient neuromorphic learning for clinical edge intelligence
- Cross-modal self-supervised learning for Alzheimer's and Parkinson's analysis

SUBMISSION PROCEDURE:

Researchers and practitioners are invited to submit papers for this particular theme session on **[session name] on or before [30th August 2026]**. All submissions must be original and may not be under review by another publication. INTERESTED AUTHORS SHOULD CONSULT THE CONFERENCE'S GUIDELINES FOR MANUSCRIPT SUBMISSIONS at <https://www.icain-conf.com/downloads>. All submitted papers will be reviewed on a double-blind, peer-review basis.

NOTE: While submitting a paper in this special session, please specify [Session Name] at the top (above paper title) of the first page of your paper.

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